

GATE CSE — 50-Question Practice Test Sheet

Original NovaTutor practice • PYQ-pattern style • Answer key and explanations included

Q1. Engineering Mathematics • Recurrences • 2 marks

Let $a_0=3$ and $a_k=2a_{(k-1)}+3$ for $k \geq 1$. What is a_4 ?

Answer type: Numerical Answer Type (NAT)

Q2. Engineering Mathematics • Recurrences • 2 marks

Let $a_0=4$ and $a_k=2a_{(k-1)}+3$ for $k \geq 1$. What is a_5 ?

Answer type: Numerical Answer Type (NAT)

Q3. Engineering Mathematics • Recurrences • 2 marks

Let $a_0=5$ and $a_k=2a_{(k-1)}+3$ for $k \geq 1$. What is a_6 ?

Answer type: Numerical Answer Type (NAT)

Q4. Engineering Mathematics • Recurrences • 2 marks

Let $a_0=6$ and $a_k=2a_{(k-1)}+3$ for $k \geq 1$. What is a_7 ?

Answer type: Numerical Answer Type (NAT)

Q5. Engineering Mathematics • Recurrences • 2 marks

Let $a_0=3$ and $a_k=2a_{(k-1)}+3$ for $k \geq 1$. What is a_8 ?

Answer type: Numerical Answer Type (NAT)

Q6. Digital Logic • Boolean Algebra • 2 marks

For a 4-variable Boolean function F , suppose $F=0$ exactly on 1 input combinations and 1 on all others. How many minterms belong to F' ?

- A. 8
- B. 16
- C. 15
- D. 1

Q7. Digital Logic • Boolean Algebra • 2 marks

For a 5-variable Boolean function F , suppose $F=0$ exactly on 2 input combinations and 1 on all others. How many minterms belong to F' ?

- A. 16
- B. 2
- C. 30
- D. 32

Q8. Digital Logic • Boolean Algebra • 2 marks

For a 6-variable Boolean function F , suppose $F=0$ exactly on 3 input combinations and 1 on all others. How many minterms belong to F' ?

- A. 64

B. 61

C. 3

D. 32

Q9. Digital Logic • Boolean Algebra • 2 marks

For a 7-variable Boolean function F , suppose $F=0$ exactly on 4 input combinations and 1 on all others. How many minterms belong to F' ?

A. 128

B. 4

C. 124

D. 64

Q10. Digital Logic • Boolean Algebra • 2 marks

For a 4-variable Boolean function F , suppose $F=0$ exactly on 1 input combinations and 1 on all others. How many minterms belong to F' ?

A. 15

B. 8

C. 1

D. 16

Q11. Computer Organization & Architecture • Cache • 2 marks

A cache has hit time 1 cycle, miss rate 0.1, and miss penalty 8 cycles. What is the average memory access time in cycles?

Answer type: Numerical Answer Type (NAT)

Q12. Computer Organization & Architecture • Cache • 2 marks

A cache has hit time 1 cycle, miss rate 0.2, and miss penalty 9 cycles. What is the average memory access time in cycles?

Answer type: Numerical Answer Type (NAT)

Q13. Computer Organization & Architecture • Cache • 2 marks

A cache has hit time 1 cycle, miss rate 0.3, and miss penalty 10 cycles. What is the average memory access time in cycles?

Answer type: Numerical Answer Type (NAT)

Q14. Computer Organization & Architecture • Cache • 2 marks

A cache has hit time 1 cycle, miss rate 0.4, and miss penalty 11 cycles. What is the average memory access time in cycles?

Answer type: Numerical Answer Type (NAT)

Q15. Computer Organization & Architecture • Cache • 2 marks

A cache has hit time 1 cycle, miss rate 0.5, and miss penalty 12 cycles. What is the average memory access time in cycles?

Answer type: Numerical Answer Type (NAT)

Q16. Programming & Data Structures • Recursion • 2 marks

A recursive procedure makes exactly one recursive call on $\text{floor}(n/2)$ and performs $\Theta(1)$ work otherwise. For $n=16$, which asymptotic bound is correct?

- A. $\Theta(n \log n)$
- B. $\Theta(\sqrt{n})$
- C. $\Theta(n)$
- D. $\Theta(\log n)$

Q17. Programming & Data Structures • Recursion • 2 marks

A recursive procedure makes exactly one recursive call on $\text{floor}(n/2)$ and performs $\Theta(1)$ work otherwise. For $n=19$, which asymptotic bound is correct?

- A. $\Theta(n \log n)$
- B. $\Theta(\log n)$
- C. $\Theta(n)$
- D. $\Theta(\sqrt{n})$

Q18. Programming & Data Structures • Recursion • 2 marks

A recursive procedure makes exactly one recursive call on $\text{floor}(n/2)$ and performs $\Theta(1)$ work otherwise. For $n=22$, which asymptotic bound is correct?

- A. $\Theta(n)$
- B. $\Theta(\log n)$
- C. $\Theta(n \log n)$
- D. $\Theta(\sqrt{n})$

Q19. Programming & Data Structures • Recursion • 2 marks

A recursive procedure makes exactly one recursive call on $\text{floor}(n/2)$ and performs $\Theta(1)$ work otherwise. For $n=25$, which asymptotic bound is correct?

- A. $\Theta(\log n)$
- B. $\Theta(n)$
- C. $\Theta(\sqrt{n})$
- D. $\Theta(n \log n)$

Q20. Programming & Data Structures • Recursion • 2 marks

A recursive procedure makes exactly one recursive call on $\text{floor}(n/2)$ and performs $\Theta(1)$ work otherwise. For $n=28$, which asymptotic bound is correct?

- A. $\Theta(\log n)$
- B. $\Theta(\sqrt{n})$
- C. $\Theta(n \log n)$
- D. $\Theta(n)$

Q21. Algorithms • Recurrences • 2 marks

Consider $T(n)=2T(n/2)+n \log n$. Which bound is correct for the recurrence under the Master theorem?

- A. $\Theta(\log n)$
- B. $\Theta(n \log n)$
- C. $\Theta(n)$
- D. $\Theta(n^2)$

Q22. Algorithms • Recurrences • 2 marks

Consider $T(n)=3T(n/3)+n \log n$. Which bound is correct for the recurrence under the Master theorem?

- A. $\Theta(\log n)$
- B. $\Theta(n)$
- C. $\Theta(n \log n)$
- D. $\Theta(n^2)$

Q23. Algorithms • Recurrences • 2 marks

Consider $T(n)=4T(n/4)+n \log n$. Which bound is correct for the recurrence under the Master theorem?

- A. $\Theta(n \log n)$
- B. $\Theta(\log n)$
- C. $\Theta(n)$
- D. $\Theta(n^2)$

Q24. Algorithms • Recurrences • 2 marks

Consider $T(n)=2T(n/5)+n \log n$. Which bound is correct for the recurrence under the Master theorem?

- A. $\Theta(n^{2.32193})$
- B. $\Theta(n)$
- C. $\Theta(n^2)$
- D. $\Theta(\log n)$

Q25. Algorithms • Recurrences • 2 marks

Consider $T(n)=3T(n/2)+n \log n$. Which bound is correct for the recurrence under the Master theorem?

- A. $\Theta(n^2)$
- B. $\Theta(\log n)$
- C. $\Theta(n)$
- D. $\Theta(n \log n)$

Q26. Theory of Computation • DFA/NFA • 2 marks

An NFA has 3 states. What is the maximum number of states in the equivalent DFA obtained by subset construction?

- A. 6
- B. 4
- C. 9
- D. 8

Q27. Theory of Computation • DFA/NFA • 2 marks

An NFA has 4 states. What is the maximum number of states in the equivalent DFA obtained by subset construction?

- A. 16
- B. 8
- C. 5
- D. 4

Q28. Theory of Computation • DFA/NFA • 2 marks

An NFA has 5 states. What is the maximum number of states in the equivalent DFA obtained by subset construction?

- A. 10
- B. 6
- C. 32
- D. 25

Q29. Theory of Computation • DFA/NFA • 2 marks

An NFA has 6 states. What is the maximum number of states in the equivalent DFA obtained by subset construction?

- A. 64
- B. 12
- C. 36
- D. 7

Q30. Theory of Computation • DFA/NFA • 2 marks

An NFA has 7 states. What is the maximum number of states in the equivalent DFA obtained by subset construction?

- A. 49
- B. 14
- C. 8
- D. 128

Q31. Compiler Design • Parsing • 2 marks

For a grammar with terminals and nonterminals, an LL(1) parser requires which condition at a choice point?

- A. LR parsing must be used
- B. the predictive choices must be uniquely determined by one lookahead symbol
- C. the grammar must have ambiguity
- D. all productions must be left recursive

Q32. Compiler Design • Parsing • 2 marks

For a grammar with terminals and nonterminals, an LL(1) parser requires which condition at a choice point?

- A. the predictive choices must be uniquely determined by one lookahead symbol
- B. all productions must be left recursive
- C. LR parsing must be used

D. the grammar must have ambiguity

Q33. Compiler Design • Parsing • 2 marks

For a grammar with terminals and nonterminals, an LL(1) parser requires which condition at a choice point?

- A. all productions must be left recursive
- B. the predictive choices must be uniquely determined by one lookahead symbol
- C. the grammar must have ambiguity
- D. LR parsing must be used

Q34. Compiler Design • Parsing • 2 marks

For a grammar with terminals and nonterminals, an LL(1) parser requires which condition at a choice point?

- A. all productions must be left recursive
- B. LR parsing must be used
- C. the grammar must have ambiguity
- D. the predictive choices must be uniquely determined by one lookahead symbol

Q35. Compiler Design • Parsing • 2 marks

For a grammar with terminals and nonterminals, an LL(1) parser requires which condition at a choice point?

- A. all productions must be left recursive
- B. the grammar must have ambiguity
- C. the predictive choices must be uniquely determined by one lookahead symbol
- D. LR parsing must be used

Q36. Operating Systems • Scheduling • 2 marks

Under round-robin scheduling with time quantum 3, a single CPU-bound process requiring 5 CPU bursts of one quantum each is interleaved with at least one other runnable process. Which qualitative effect occurs as q decreases?

- A. context-switch overhead increases
- B. the process never waits
- C. deadlock becomes impossible
- D. turnaround time must always decrease

Q37. Operating Systems • Scheduling • 2 marks

Under round-robin scheduling with time quantum 4, a single CPU-bound process requiring 6 CPU bursts of one quantum each is interleaved with at least one other runnable process. Which qualitative effect occurs as q decreases?

- A. deadlock becomes impossible
- B. turnaround time must always decrease
- C. the process never waits
- D. context-switch overhead increases

Q38. Operating Systems • Scheduling • 2 marks

Under round-robin scheduling with time quantum 5, a single CPU-bound process requiring 7 CPU bursts of one quantum each is interleaved with at least one other runnable process. Which qualitative effect occurs as q decreases?

- A. the process never waits
- B. turnaround time must always decrease
- C. context-switch overhead increases
- D. deadlock becomes impossible

Q39. Operating Systems • Scheduling • 2 marks

Under round-robin scheduling with time quantum 6, a single CPU-bound process requiring 8 CPU bursts of one quantum each is interleaved with at least one other runnable process. Which qualitative effect occurs as q decreases?

- A. deadlock becomes impossible
- B. context-switch overhead increases
- C. turnaround time must always decrease
- D. the process never waits

Q40. Operating Systems • Scheduling • 2 marks

Under round-robin scheduling with time quantum 3, a single CPU-bound process requiring 9 CPU bursts of one quantum each is interleaved with at least one other runnable process. Which qualitative effect occurs as q decreases?

- A. turnaround time must always decrease
- B. deadlock becomes impossible
- C. context-switch overhead increases
- D. the process never waits

Q41. Databases • Functional Dependencies • 2 marks

A relation has attributes A,B,C,D and functional dependencies $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$. What is the closure A^+ ?

- A. {A}
- B. {A,B}
- C. {A,B,C,D}
- D. {A,D}

Q42. Databases • Functional Dependencies • 2 marks

A relation has attributes A,B,C,D and functional dependencies $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$. What is the closure A^+ ?

- A. {A,D}
- B. {A,B}
- C. {A,B,C,D}
- D. {A}

Q43. Databases • Functional Dependencies • 2 marks

A relation has attributes A,B,C,D and functional dependencies $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$. What is the closure A^+ ?

- A. {A,B}
- B. {A,B,C,D}
- C. {A,D}
- D. {A}

Q44. Databases • Functional Dependencies • 2 marks

A relation has attributes A,B,C,D and functional dependencies $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$. What is the closure A^+ ?

- A. {A,B}
- B. {A,D}
- C. {A,B,C,D}
- D. {A}

Q45. Databases • Functional Dependencies • 2 marks

A relation has attributes A,B,C,D and functional dependencies $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$. What is the closure A^+ ?

- A. {A,B,C,D}
- B. {A,D}
- C. {A,B}
- D. {A}

Q46. Computer Networks • Subnetting • 2 marks

For IPv4 subnet /24, how many usable host addresses are available per subnet?

Answer type: Numerical Answer Type (NAT)

Q47. Computer Networks • Subnetting • 2 marks

For IPv4 subnet /23, how many usable host addresses are available per subnet?

Answer type: Numerical Answer Type (NAT)

Q48. Computer Networks • Subnetting • 2 marks

For IPv4 subnet /22, how many usable host addresses are available per subnet?

Answer type: Numerical Answer Type (NAT)

Q49. Computer Networks • Subnetting • 2 marks

For IPv4 subnet /21, how many usable host addresses are available per subnet?

Answer type: Numerical Answer Type (NAT)

Q50. Computer Networks • Subnetting • 2 marks

For IPv4 subnet /24, how many usable host addresses are available per subnet?

Answer type: Numerical Answer Type (NAT)

Answer Key & Explanations

Q1. Answer: 93

Solve the affine recurrence: $a_{k+3}=2(a_{(k-1)+3})$, hence $a_n=(a_0+3)2^n-3$.

Q2. Answer: 221

Solve the affine recurrence: $a_{k+3}=2(a_{(k-1)+3})$, hence $a_n=(a_0+3)2^n-3$.

Q3. Answer: 509

Solve the affine recurrence: $a_{k+3}=2(a_{(k-1)+3})$, hence $a_n=(a_0+3)2^n-3$.

Q4. Answer: 1149

Solve the affine recurrence: $a_{k+3}=2(a_{(k-1)+3})$, hence $a_n=(a_0+3)2^n-3$.

Q5. Answer: 1533

Solve the affine recurrence: $a_{k+3}=2(a_{(k-1)+3})$, hence $a_n=(a_0+3)2^n-3$.

Q6. Answer: 1

F' is 1 exactly where F is 0, so its minterm count equals the number of zero combinations of F .

Q7. Answer: 2

F' is 1 exactly where F is 0, so its minterm count equals the number of zero combinations of F .

Q8. Answer: 3

F' is 1 exactly where F is 0, so its minterm count equals the number of zero combinations of F .

Q9. Answer: 4

F' is 1 exactly where F is 0, so its minterm count equals the number of zero combinations of F .

Q10. Answer: 1

F' is 1 exactly where F is 0, so its minterm count equals the number of zero combinations of F .

Q11. Answer: 1.8

$AMAT = \text{hit time} + \text{miss rate} \times \text{miss penalty}$.

Q12. Answer: 2.8

$AMAT = \text{hit time} + \text{miss rate} \times \text{miss penalty}$.

Q13. Answer: 4.0

$AMAT = \text{hit time} + \text{miss rate} \times \text{miss penalty}$.

Q14. Answer: 5.4

$AMAT = \text{hit time} + \text{miss rate} \times \text{miss penalty}$.

Q15. Answer: 7.0

$AMAT = \text{hit time} + \text{miss rate} \times \text{miss penalty}$.

Q16. Answer: $\Theta(\log n)$

Each call halves the input, giving a logarithmic recursion depth.

Q17. Answer: $\Theta(\log n)$

Each call halves the input, giving a logarithmic recursion depth.

Q18. Answer: $\Theta(\log n)$

Each call halves the input, giving a logarithmic recursion depth.

Q19. Answer: $\Theta(\log n)$

Each call halves the input, giving a logarithmic recursion depth.

Q20. Answer: $\Theta(\log n)$

Each call halves the input, giving a logarithmic recursion depth.

Q21. Answer: $\Theta(n \log n)$

Compare $f(n) = n \log n$ with $n^{(\log_b a)}$. The correct case depends on the comparison; the generated instance is chosen so the displayed answer is the dominating term.

Q22. Answer: $\Theta(n \log n)$

Compare $f(n) = n \log n$ with $n^{(\log_b a)}$. The correct case depends on the comparison; the generated instance is chosen so the displayed answer is the dominating term.

Q23. Answer: $\Theta(n \log n)$

Compare $f(n) = n \log n$ with $n^{(\log_b a)}$. The correct case depends on the comparison; the generated instance is chosen so the displayed answer is the dominating term.

Q24. Answer: $\Theta(n^{2.32193})$

Compare $f(n) = n \log n$ with $n^{(\log_b a)}$. The correct case depends on the comparison; the generated instance is chosen so the displayed answer is the dominating term.

Q25. Answer: $\Theta(n \log n)$

Compare $f(n) = n \log n$ with $n^{(\log_b a)}$. The correct case depends on the comparison; the generated instance is chosen so the displayed answer is the dominating term.

Q26. Answer: 8

Each DFA state corresponds to a subset of NFA states, giving at most 2^n subsets.

Q27. Answer: 16

Each DFA state corresponds to a subset of NFA states, giving at most 2^n subsets.

Q28. Answer: 32

Each DFA state corresponds to a subset of NFA states, giving at most 2^n subsets.

Q29. Answer: 64

Each DFA state corresponds to a subset of NFA states, giving at most 2^n subsets.

Q30. Answer: 128

Each DFA state corresponds to a subset of NFA states, giving at most 2^n subsets.

Q31. Answer: the predictive choices must be uniquely determined by one lookahead symbol

LL(1) means one-token lookahead is sufficient to choose the production without conflict.

Q32. Answer: the predictive choices must be uniquely determined by one lookahead symbol

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LL(1) means one-token lookahead is sufficient to choose the production without conflict.

Q35. Answer: the predictive choices must be uniquely determined by one lookahead symbol

LL(1) means one-token lookahead is sufficient to choose the production without conflict.

Q36. Answer: context-switch overhead increases

A smaller quantum causes more preemptions and therefore more context-switch overhead.

Q37. Answer: context-switch overhead increases

A smaller quantum causes more preemptions and therefore more context-switch overhead.

Q38. Answer: context-switch overhead increases

A smaller quantum causes more preemptions and therefore more context-switch overhead.

Q39. Answer: context-switch overhead increases

A smaller quantum causes more preemptions and therefore more context-switch overhead.

Q40. Answer: context-switch overhead increases

A smaller quantum causes more preemptions and therefore more context-switch overhead.

Q41. Answer: {A,B,C,D}

Starting with A, repeatedly apply A->B, then B->C, then C->D.

Q42. Answer: {A,B,C,D}

Starting with A, repeatedly apply A->B, then B->C, then C->D.

Q43. Answer: {A,B,C,D}

Starting with A, repeatedly apply A->B, then B->C, then C->D.

Q44. Answer: {A,B,C,D}

Starting with A, repeatedly apply A->B, then B->C, then C->D.

Q45. Answer: {A,B,C,D}

Starting with A, repeatedly apply A->B, then B->C, then C->D.

Q46. Answer: 254

Usable hosts = $2^{(32-\text{prefix})}-2$ for conventional IPv4 host addressing.

Q47. Answer: 510

Usable hosts = $2^{(32-\text{prefix})}-2$ for conventional IPv4 host addressing.

Q48. Answer: 1022

Usable hosts = $2^{(32-\text{prefix})}-2$ for conventional IPv4 host addressing.

Q49. Answer: 2046

Usable hosts = $2^{(32-\text{prefix})}-2$ for conventional IPv4 host addressing.

Q50. Answer: 254

Usable hosts = $2^{(32-\text{prefix})}-2$ for conventional IPv4 host addressing.